



Question: If $x = 7\sqrt{3}$, find a .

Correct Answer: $14\sqrt{3}$

MCTS

Image Description:

The image shows triangle ABC with a right angle at vertex C. Inside the triangle, **a perpendicular line segment BE drops from point B to side AC**, and another perpendicular CE drops to side AB. Angle B = 60° and angle A = 30° . The segment $BE = x = 7\sqrt{3}$, and we are to find the length of side $a = BC$.

[Perception Error] A single model trapped in a wrong image description.

Rationales:

In right triangle BEC, angle CBE = 60° and **BE is perpendicular to AC**, making triangle BEC a right triangle with a known angle and opposite side. We can use the sine function:

$$\sin(60^\circ) = \text{opposite} / \text{hypotenuse} = x / a$$

Let's think step by step.

Step 1:

Use the sine ratio in triangle BEC:

$$\sin(60^\circ) = x / a$$

[Reasoning Error] Ultimately leads to incorrect reasoning.

Step 2:

Substitute the known values:

$$\sin(60^\circ) = 7\sqrt{3} / a, \text{ and } \sin(60^\circ) = \sqrt{3} / 2$$

Step 3:

Solve the equation:

$$\sqrt{3} / 2 = 7\sqrt{3} / a$$

Step 4:

Cross multiply:

$$\sqrt{3} / 2 = 7\sqrt{3} / a \rightarrow a = (7\sqrt{3} \times 2) / \sqrt{3} = 14$$

The final answer is: **14**.

CoMCTS:

Image Description:

The diagram shows a right triangle ABC with right angle at vertex C. Angle B = 60° , angle A = 30° . **A perpendicular segment CE is drawn from point C to side AB**, forming a right angle at point E. The length of BE is given as $x = 7\sqrt{3}$. Side BC = a is to be found.

Rationales:

We are dealing with right triangle BEC, where angle CBE = 60° , **BE = x is the leg adjacent to the angle, and BC = a is the hypotenuse**.

Therefore, the cosine function is appropriate:

$$\cos(\theta) = \text{adjacent} / \text{hypotenuse}$$

$$\Rightarrow \cos(60^\circ) = x / a$$

Let's think step by step.

Step 1:

Identify the known values:

$$- x = 7\sqrt{3}$$

$$- \cos(60^\circ) = 1/2$$

Multiple models leverage collective perceptual and reasoning abilities to ultimately arrive at the correct answer.

Step 2:

Use the cosine definition in triangle BEC:

$$\cos(60^\circ) = x / a$$

$$\Rightarrow 1/2 = 7\sqrt{3} / a$$

Step 3:

Solve for a by cross-multiplying:

$$a = 2 \times 7\sqrt{3} = 14\sqrt{3}$$

The final answer is: **$14\sqrt{3}$** .